

Curriculum Vitæ

Alexandre BOUSSE

1 Personal Details

First Name:	Alexandre
Family Name:	BOUSSE
Date of Birth:	13th of June, 1980
Place of Birth:	Rennes, France
Citizenship:	French
Position:	Full Researcher (<i>chargé de recherche</i>)
Laboratory:	Laboratoire de Traitement de l'Information Médicale (LaTIM) Institut national de la santé et de la recherche médicale (Inserm), UMR 1101, Brest, France
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2 Academic Career

2.1 Professional Experience

2021–present	Full Researcher (<i>chargé de recherche</i>), LaTIM, Inserm, UMR 1101, Université de Bretagne Occidentale (UBO), Brest, France
2018–2021	Associate Professor , LaTIM, INSERM, UMR 1101, UBO, Brest, France
2009–2018	Post-doctorate , Institute of Nuclear Medicine (INM), University College London (UCL), London, UK
2005–2008	PhD Candidate , Laboratoire du Traitement du Signal et de l'Image (LTSI), Inserm, UMR 1099, <i>Université de Rennes 1</i> , Rennes, France

2.2 Education

2019	Habilitation à diriger des recherches (HDR) (research habilitation thesis), LaTIM, INSERM, UMR 1101, UBO, Brest, France Title: “Contributions à la reconstruction tomographique compensée en mouvement” Viva: 07/10/2019
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	Jury: • Dimitris Visvikis, LaTIM, Inserm, UMR 1101, UBO, Brest, France • Françoise Pène, UBO, Brest, France • Claude Comtat, Commissariat à l'Énergie Atomique et aux énergies alternatives (CEA), Orsay, France • Andrew Reader, King's College London (KCL), London, UK • Michel Defrise, Université Libre de Bruxelles, Brussels, Belgium
2005–2008	PhD, Signal Processing, LTSI, UMR 1099, Université de Rennes 1, Rennes, France, and Laboratory of Image Science and Technology (LIST), Southeast University (SEU), Nanjing, China Title: “Inverse Problems and Application to Motion-Compensated Rotational X-ray Angiography” Viva: 08/12/2008 Jury: • Directors: – J.-L. Coatrieux, LTSI, Inserm, UMR 1099, Université de Rennes 1, Rennes, France – H. Shu, LIST, SEU, Nanjing, China – C. Toumoulin, LTSI, Inserm, UMR 1099, Université de Rennes 1, Rennes, France • President: J. Demongeot, Université Joseph Fourier, Grenoble, France • Referees: – J. Yang, Nanjing University of Science and Technology, Nanjing, China – C. Roux, Télécom Bretagne, Brest, France • Reviewers: – L. Luo, LIST, SEU, Nanjing, China – D. Xia, Nanjing University of Science and Technology, Nanjing, China
2004–2005	Master of Science (DEA), Statistics , Université de Rennes 1, Rennes, France
2003–2004	Master of Science (DESS), Statistics , Université de Rennes 1, Rennes, France
1998–2003	Bachelor of Science, Mathematics , Université de Rennes 1, Rennes, France

3 Research Activities

3.1 PhD Supervision and Co-supervision

2024–present	Clémentine Phung-Ngoc—UBO Title: “Motion-corrected Low-dose PET/CT with Unsupervised Learning” Director: Alexandre Bousse (50%) Co-director: Olivier Saut (25%), Hong-Phuong Dang (25%) Publications: [PP1], [CP3] Status: ongoing
2024–present	Antoine De Paepe—UBO Title: “Motion-compensated CT Reconstruction with Unsupervised Learning” Director: Alexandre Bousse (100%) Publications: [J4], [CP2] Status: ongoing

2023–present	Thore Dassow —UBO Title: “Scored-Based Diffusion Models for Material Decomposition in PCCT” Director: Alexandre Bousse (100%) Publications: [CP1] (best Fully3D 2025 paper award) Status: ongoing
2022–present	Corentin Vazia —Université Bretagne Sud (UBS) Title: “Une approche de la reconstruction d’images et de la décomposition de matériaux en tomodensitométrie spectrale avec régularisation par modèle de diffusion” Director: Jacques Froment (50%) Co-director: Alexandre Bousse (50%) Publications: [J2], [CP9], [CP10] Status: ongoing
2021–2025	Juan José Molina —Pontificia Universidad Católica de Chile (PUC) Title: “Deep Learning Techniques for T_2^* Image Reconstruction” Director: Matias Courdurier (80%) Co-supervisor: Alexandre Bousse (20%) Publications: [PP3] Status: completed
2021–2025	Youness Mellak —UBO Title: “Approches par apprentissage profond pour l’imagerie TEP : TEP à trois photons, correction de la portée du positron et reconstruction directe” Director: Dimitris Visvikis (25%) Co-director: Alexandre Bousse (75%) Publications: [J1], [J6], [CP6]–[CP8], [CP16] Status: completed
2021–2025	Valentin Gautier —Institut National des Sciences Appliquées (INSA) Lyon Title: “ <i>Reconstruction bimodale d’images TEP/IRM assistée par intelligence artificielle</i> ” Director: Bruno Sixou (50%) Co-supervisor: Alexandre Bousse (25%), Voichita Maxim (25%) Publications: [J8], [CP5], [CP12] Status: completed
2021–2024	Noel Pinton —UBO Title: “Synergistic PET/CT Reconstruction using Deep Learning” Director: Alexandre Bousse (100%) Publications: [J7], [CP13], [CP14] Status: completed
2021–2024	Zhihan Wang —UBO Title: “Spectral computed tomographic image reconstruction using deep learning” Director: Alexandre Bousse (75%) Co-director: Frank Vermet (25%) Publications: [CP4], [J9], [CP17] Status: completed
2017–2022	Baptiste Laurent —UBO Title: “ <i>Estimation des diffusés en TEP par apprentissage profond</i> ” Director: Nicolas Boussion Co-director: Alexandre Bousse (100%) Publications: [J5], [J10], [CP23]

	Status: completed
2018–2022	Sai Sundar Kandarpa —UBO Title: “Tomographic Image Reconstruction with Direct Neural Network Approaches” Director: Alexandre Bousse (100%) Publications: [RV1], [RV2], [CP15], [J11], [J15], [CP20], [CP29] Status: completed
2018–2022	Suxer Alfonso Garcia —UBO Title: “Multi-channel Computed Tomographic Image Reconstruction by Exploiting Structural Similarities” Director: Alexandre Bousse (100%) Publications: [J12], [CP18], [CP19], [CP21] Status: completed
2017–2020	Debora Giovagnoli —Institut Mines-Télécom (IMT) Title: “3- γ Image Reconstruction using LXe Compton Camera XEMIS2” Director: Dimitris Visvikis Co-director: Alexandre Bousse (50%) Publications: [J14] Status: completed
2016–2020	Ludovica Brusafferri —UCL Title: “Improving Quantification in non-TOF 3D PET/MR by Incorporating Photon Energy Information” Director: Kris Thielemans (50%) Co-supervisor: Alexandre Bousse (50%) Publications: [J13], [J18], [CP22], [CP32], [CP35] Status: completed
2016–2020	Élise Émond —UCL Title: “Improving Quantification in Lung PET/CT for the Evaluation of Disease Progression and Treatment Effectiveness” Director: Kris Thielemans (50%) Co-supervisor: Alexandre Bousse (50%) Publications: [J19], [J20], [CP25]–[CP27] Status: completed
2014–2018	Yu-Jung Tsai —UCL Title: “Penalised Image Reconstruction Algorithms for Efficient and Consistent Quantification in Emission Tomography” Director: Kris Thielemans (50%) Co-supervisor: Alexandre Bousse (50%) Publications: [J16], [J21], [CP33], [J23], [CP36], [CP39], [CP42] Status: completed
2010–2015	Sarah Cade —UCL Title: “Attenuation Correction of Myocardial Perfusion Scintigraphy Images without Transmission Scanning” Director: Brian F. Hutton (50%) Co-supervisor: Alexandre Bousse (50%) Publications: [CP58] Status: completed

3.2 Master Students Supervision

2026	Peifeng Wang (École d'Ingénieur Statistique Data Science et Big Data (ENSAI)) "Foundation models for PCCT reconstruction"
2024	Antoine De Paepe, Clementine Phung-Ngoc, Apolline Guerineau (ENSAI) "Compressed-sensing for CT reconstruction" (projet de fin d'études (PFE))
2023	Tiban Dorel (IMT Atlantique) "CT reconstruction with unrolling architectures"
2022	Mariana Yuli Sato do Nascimento (IMT Atlantique) "PET/CT denoising with conditional GANs"
2019	Celia Boutalbi (Université de Bordeaux) "Direct material decomposition in dual-energy CT"

3.3 Teachings

2025–present	ENSAI , Bruz, France Machine and deep learning 9-hour lecture per year.
2018–present	IMT Atlantique (2nd year), Brest, France Advanced Medical Image Reconstruction 5-hour lecture per year & 6-hour lab per year Mathematics and algorithms for image reconstruction Machine and deep learning.
2018–2021	M2 Master Signaux Images en Biologie et Médecine (SIBM) , UBO, Brest, France Introduction to Image Reconstruction 6-hour lecture per year (ended in 2021) France: 12-hour lecture & 6-hour lab per year
2018–2022	M2 <i>physique et instrumentation</i> , UBO, Brest Introduction to Image Reconstruction 12-hour lecture & 6-hour lab per year. (ended in 2022)
2018–2021	Coordinator of M2 SIBM at UBO
2018–2021	M1 biologie-santé , UBO, Brest, France Image Segmentation 6-hours lecture per year.
2018–2021	Medical and Dental School , UBO, Brest, France Pix 100 hours lab per year.
2004–2007	ENSAI (first year), Bruz, France Probability & Statistics (contractual teaching during PhD) Teaching topics: measure theory, random variables, parametric statistics, statistical hypothesis testing, linear regression 64-hour science lab in total.

4 Grants and External Funding

4.1 Funding at Current Position (since 01/09/2018)

2025–present	Projets Intra- et Inter-Instituts Brestois (IB) de l’UBO (France) Amount awarded: €6,000 Role: PI Project Title: “Diffusion Spectrale”
2024–present	Inserm-Inria PhD grant (France) Amount awarded: €130,000 Role: PI Project Title: “Motion-corrected Low-dose PET/CT with Unsupervised Learning”
2024–present	Région Bretagne / UBO PhD grant (France) Amount awarded: €130,000 Role: PI Project Title: “Motion-compensated CT Reconstruction with Unsupervised Learning”
2023–present	France Life Imaging WP4 (France) Amount awarded: €20,000 Role: PI Project Title: “Deepmulti: Deep Learning Techniques for Multimodal Image Reconstruction”
2020–2024	ANR – ANR-20-CE45-0020 (France) Amount awarded: €496,800 Role: PI Project Title: “MultiRecon: Machine-Learning for Multimodal Medical Image Reconstruction”
2019–2022	France Life Imaging WP4 (France) Amount awarded: €24,000 Role: PI Project Title: “Dual-Tracer in Dynamic PET” (complement funding to the <i>Émergence</i> project)
2019–2022	AO <i>Émergence Cancéropôle Grand Ouest</i> (France) Amount awarded: €15,000 Role: PI Project Title: “Dual-Tracer in Dynamic PET”

4.2 Funding during Postdoc at UCL (2009–2018)

2016–2018	GE Healthcare (USA) Amount awarded: \$150,000 Role: co-PI PI: Kris Thielemans Project Title: “Motion-Compensated PET/CT”
2013–2015	FP7 – HEALTH Program 305311 (EU) Amount awarded: €575,622 (€5,981,463 in total for all the EU partners) Role: Research Fellow PI: Brian F. Hutton Project Title: “Development of an integrated SPECT/MRI system”

2013–2016	EPSRC – EP/K005278/1 (UK) Amount awarded: £1,274,298 Role: WP leader PI: Brian F. Hutton Project Title: “Exploiting the Unique Quantitative Capabilities Offered by Simultaneous PET/MRI”
2009–2012	EPSRC – EP/G026483/1 (UK) Amount awarded: £767,088 Role: Research Fellow PI: Brian F. Hutton Project Title: “Optimising Reconstruction to Accommodate Complex System Models for Emission Tomography”

5 Academic Service and Scientific Diffusion

5.1 Meetings Organisation

5.1.1 Conferences

2022–present	Member of IEEE Nuclear and Medical Imaging Sciences Council
2023	Session Chairman , IEEE Medical Imaging Conference, Vancouver, Canada
2022	Session Chairman , IEEE Medical Imaging Conference, Milan, Italy
2021	Student Paper Awards Committee , International Conference on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine
2018	Session Chairman , IEEE Medical Imaging Conference, Sydney, Australia

5.1.2 Workshops

2022	AI Wild West (organiser) https://wildwestworkshop.github.io/
2022	Rencontre Lyonnaise en Imagerie d’Émission (organiser) https://emilyworkshop.github.io/

5.2 Scientific Evaluation

5.2.1 Reviewing

Journal Associate Editor

- IEEE Transactions on Radiation and Plasma Medical Sciences

Journal Peer Review

- IEEE Transactions on Medical Imaging
- IEEE Transactions on Biomedical Engineering
- IEEE Transactions on Radiation and Plasma Medical Sciences
- IEEE Transactions on Computational Imaging
- IEEE Transactions on Consumer Electronics
- Physics in Medicine & Biology

- Neuroimage
- PLOS one
- Philosophical Transactions A
- Frontiers in Nuclear Medicine
- European Journal of Nuclear Medicine and Molecular Imaging

Grant Peer Review

- Nantes Excellence Trajectory (NEXT)
- *Wetenschappelijk Fonds Willy Gepts* (WFWG)
- Netherlands Organisation for Scientific Research (NWO)
- Expertise Scientifique Cifre

Conference Committee

- MICCAI 2020, 2021, 2022, 2023, 2024, 2025, 2026
- Fully 3D 2021, 2023, 2025
- IEEE Nuclear Science Symposium and Medical Imaging Conference 2014, 2015, 2016, 2017, 2018, 2019, 2021, 2022, 2023, 2024, 2025, 2026

5.2.2 PhD Examination

2023	Kannara Mom , “Deep learning-based phase retrieval for X-ray phase contrast imaging”, INSA Lyon, Lyon, France.
2023	Guillaume Corda , “Joint PET-MR deep learned reconstruction”, King’s College London, London, UK.
2022	Louise Friot , “ <i>Méthodes de reconstruction avancées en tomographie dentaire par faisceau conique</i> ”, INSA Lyon, Lyon, France.
2021	Zacharias Chalampalak , “Modelling and Reconstruction of Whole-Body Parametric Maps in PET-MRI Pharmacological Imaging”, Biomaps, Orsay, France.

5.2.3 PhD Jury

2026	Mehdi Latif , “Reconstruction tomographique pour l’imagerie 3-photons basée sur des caméras hybrides TEP/Compton”, Centrale Nantes, Nantes, France.
2025	Corentin Vazia , “Une approche de la reconstruction d’images et de la décomposition de matériaux en tomodensitométrie spectrale avec régularisation par modèle de diffusion”, UBS, Vannes, France.
2025	Youness Mellak , “Approches par apprentissage profond pour l’imagerie TEP : TEP à trois photons, correction de la portée du positron et reconstruction directe”, UBO, Brest, France.
2024	Noel Pinton , “Synergistic PET/CT Reconstruction using Deep Learning”, UBO, Brest, France.
2024	Zhihan Wang , “Spectral computed tomographic image reconstruction using deep learning”, UBO, Brest, France.
2022	Suxer Alfonso Garcia , “Multi-channel computed tomographic image reconstruction by exploiting structural similarities”, UBO, Brest, France.
2022	Sai Sundar Kandarpa , “Tomographic image reconstruction with direct neural network approaches”, UBO, Brest, France.
2020	Debora Giovagnoli , “3- γ Image Reconstruction using LXe Compton Camera XEMIS2”, UBO, Brest, France.

5.2.4 CSIs

2025–2027	Cyprien Lanneau , “Estimation de mouvements non rigides à partir de projections CBCT”, Paris Saclay, Paris, France.
2023–2025	Vuk Ignjatovic , “Motion Correction in Interventional 3D X-Ray Stroke Imaging”, CentraleSupélec, Paris, France
2023–2025	Younes Moussaoui , “Approches neuronales profondes pour l’estimation des paramètres de régularisation et la représentation d’images en reconstruction TEP”, Centrale Nantes, Nantes, France.
2022–2024	Alexandre Merasli , “Reconstruction d’images TEP par des méthodes d’optimisation hybrides utilisant un réseau de neurones non supervisé et de l’information anatomique”, Siemens Healthineers

5.2.5 Other Jurys

2022–present	Jury contrat doctoral établissement (CDE), école doctorale biologie santé, UBO.
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5.3 Notable Invited Talks

Mar. 2026	“Unsupervised Learning for CT and PET Reconstruction with Generative Models”, Synerbi 2026, London, UK.
Oct. 2025	“Deep Learning for Multimodal Image Reconstruction”, CEA, Paris, France.
Jul. 2024	“Tomodensitométrie à comptage photonique : projets du LaTIM en reconstruction d’image”, LaTIM, Brest, France.
Nov. 2023	“Modèles génératifs pour reconstruction multimodale”, GDR ISIS, INSA, Lyon, France.
Mar. 2022	“Generative Models for Multichannel Image Reconstruction”,

	<i>Pontificia Universidad Católica de Chile</i> , Santiago, Chile.
Oct. 2021	“Multirecon: Machine Learning for Multimodal Medical Image Reconstruction”, INSA, Lyon, France.
Jul. 2019	“Innovations in Image Reconstruction”, LaTIM, Brest, France.
Feb. 2019	“ <i>Reconstruction d’image en tomographie à émission de positons par maximum de vraisemblance avec compensation du mouvement respiratoire</i> ”, <i>Laboratoire de Mathématiques de Bretagne Atlantique</i> , Brest, France.
Jun. 2018	“Respiratory Motion Correction in PET/CT and PET/MR”, Mathematical Methods for Spatiotemporal Imaging, SIAM Conference on Imaging Science 2018, Bologna, Italy.
Mar. 2017	“Maximum-Likelihood PET Reconstruction and Motion Estimation”, <i>Pontificia Universidad Católica de Chile</i> , Santiago, Chile.
Sept. 2016	“Direct Motion Compensation in Attenuation-Corrected PET/CT and PET/MR Reconstruction”, UCL PET/MR Methods Symposium, London, UK.
May 2016	“ <i>Reconstruction en PET-CT avec compensation du mouvement par techniques de maximum de vraisemblance</i> ”, CEA, Orsay, France.
Mar. 2016	“Motion-Compensated PET Image Reconstruction by Maximum-Likelihood”, Newton Project Workshop on Brazil/UK Collaboration: the Future of Molecular Imaging, Recife, Brazil.
Nov. 2015	“Gated PET Reconstruction with Motion Compensation and Attenuation Correction using non-Gated CT”, Brain Institute, <i>Hospital Israelita Albert Einstein</i> , São Paulo, Brazil.

6 Publications

Journal Articles

- [J1] Y. Mellak, **A. Bousse**, T. Merlin, D. Giovagnoli, and D. Visvikis, “Direct3 γ pet: A pipeline for direct three-gamma PET image reconstruction,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 10, no. 2, pp. 181–191, 2026. DOI: [10.1109/TRPMS.2025.3577810](https://doi.org/10.1109/TRPMS.2025.3577810). [Online]. Available: <https://arxiv.org/abs/2407.18337>.
- [J2] C. Vazia, T. Dassow, **A. Bousse**, J. Froment, B. Vedel, F. Vermet, A. Perelli, J.-P. Tasu, and D. Visvikis, “Material decomposition in photon-counting computed tomography with diffusion models: Comparative study and hybridization with variational regularizers,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, 2026. DOI: [10.1109/TRPMS.2026.3651354](https://doi.org/10.1109/TRPMS.2026.3651354). [Online]. Available: <https://arxiv.org/abs/2503.15383>.
- [J3] J. Zhang, **A. Bousse**, L. Imbert, S. Xue, K. Shi, and J. Bert, “Semi-supervised learning for dose prediction in targeted radionuclide: A synthetic data study,” *Physics in Medicine & Biology*, vol. 71, no. 5, 2026. DOI: [10.1088/1361-6560/ae36df](https://doi.org/10.1088/1361-6560/ae36df). [Online]. Available: <https://arxiv.org/abs/2503.05367>.
- [J4] A. De Paepe, **A. Bousse**, C. Phung-Ngoc, Y. Mellak, and D. Visvikis, “Adaptive diffusion models for sparse-view motion-corrected head cone-beam CT,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, 2025. DOI: [10.1109/TRPMS.2025.3637124](https://doi.org/10.1109/TRPMS.2025.3637124). [Online]. Available: <https://arxiv.org/abs/2504.14033>.

- [J5] B. Laurent, **A. Bousse**, T. Merlin, A. Rominger, K. Shi, and D. Visvikis, “Evaluation of deep learning-based scatter correction on a long-axial field-of-view PET scanner,” *European Journal of Nuclear Medicine and Molecular Imaging*, vol. 52, no. 7, pp. 2563–2576, 2025. DOI: [10.1007/s00259-025-07120-6](https://doi.org/10.1007/s00259-025-07120-6). [Online]. Available: <https://arxiv.org/abs/2501.01341>.
- [J6] Y. Mellak, **A. Bousse**, T. Merlin, É. Émond, and D. Visvikis, “Dual-input dynamic convolution for positron range correction in PET image reconstruction,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, 2025. DOI: [10.1109/TRPMS.2025.3647264](https://doi.org/10.1109/TRPMS.2025.3647264). [Online]. Available: <https://arxiv.org/abs/2503.00587>.
- [J7] N. J. Pinton, **A. Bousse**, C. Cheze-Le Rest, and D. Visvikisc, “Multibranch generative models for multichannel imaging with an application to PET/CT synergistic reconstruction,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 9, no. 5, pp. 654–666, 2025. DOI: [10.1109/TRPMS.2025.3532176](https://doi.org/10.1109/TRPMS.2025.3532176). [Online]. Available: <https://arxiv.org/abs/2404.08748>.
- [J8] V. Gautier, **A. Bousse**, F. Sureau, C. Comtat, V. Maxim, and B. Sixou, “Bimodal PET/MRI generative reconstruction based on VAE architectures,” *Physics in Medicine & Biology*, vol. 69, 2024. DOI: [10.1088/1361-6560/ad9133](https://doi.org/10.1088/1361-6560/ad9133). [Online]. Available: <https://iopscience.iop.org/article/10.1088/1361-6560/ad9133>.
- [J9] Z. Wang, **A. Bousse**, F. Vermet, J. Froment, B. Vedel, A. Perelli, J.-P. Tasu, and D. Visvikis, “Uconnect: Synergistic spectral CT reconstruction with U-Nets connecting the energy bins,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 8, no. 2, pp. 222–233, 2024. DOI: [10.1109/TRPMS.2023.3330045](https://doi.org/10.1109/TRPMS.2023.3330045). [Online]. Available: <https://arxiv.org/abs/2311.00666>.
- [J10] B. Laurent, **A. Bousse**, T. Merlin, S. Nekolla, and D. Visvikis, “PET scatter estimation using deep learning U-net architecture,” *Physics in Medicine & Biology*, vol. 68, no. 6, p. 065 004, 2023. DOI: [10.1088/1361-6560/ac9a97](https://doi.org/10.1088/1361-6560/ac9a97).
- [J11] V. S. S. Kandarpa, A. Perelli, **A. Bousse**, and D. Visvikis, “LRR-CED: Low-resolution reconstruction-aware convolutional encoder–decoder network for direct sparse-view CT image reconstruction,” *Physics in Medicine & Biology*, vol. 67, no. 15, p. 155 007, 2022. DOI: [10.1088/1361-6560/ac7bce](https://doi.org/10.1088/1361-6560/ac7bce).
- [J12] A. Perelli, S. L. Alfonso Garcia, **A. Bousse**, J.-P. Tasu, N. Efthimiadis, and D. Visvikis, “Multi-channel convolutional analysis operator learning for dual-energy CT reconstruction,” *Physics in Medicine & Biology*, vol. 67, no. 6, p. 065 001, 2022. DOI: [10.1088/1361-6560/ac4c32](https://doi.org/10.1088/1361-6560/ac4c32). [Online]. Available: <https://arxiv.org/abs/2203.05968v1>.
- [J13] L. Brusaferrì, É. C. Émond, **A. Bousse**, R. Twyman, A. C. Whitehead, D. Atkinson, S. Ourselin, B. F. Hutton, S. Arridge, and K. Thielemans, “Detection efficiency modelling and joint activity and attenuation reconstruction in non-TOF 3D PET from multiple-energy window data,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 6, no. 1, pp. 87–97, 2021. DOI: [10.1109/TRPMS.2021.3064239](https://doi.org/10.1109/TRPMS.2021.3064239).
- [J14] D. Giovagnoli, **A. Bousse**, N. Beaupere, C. Canot, J.-P. Cussonneau, S. Diglio, A. Iborra Carreres, J. Masbou, T. Merlin, E. Morteau, Y. Xing, Y. Zhu, D. Thers, and D. Visvikis, “A pseudo-TOF image reconstruction approach for three-gamma small animal imaging,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 5, no. 6, pp. 826–834, 2021. DOI: [10.1109/TRPMS.2020.3046409](https://doi.org/10.1109/TRPMS.2020.3046409).
- [J15] V. S. S. Kandarpa, **A. Bousse**, D. Benoit, and D. Visvikis, “DUG-RECON: A framework for direct image reconstruction using convolutional generative networks,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 5, no. 1, pp. 44–53, 2021. DOI: [10.1109/TRPMS.2020.3033172](https://doi.org/10.1109/TRPMS.2020.3033172). [Online]. Available: <https://arxiv.org/abs/2012.02000>.
- [J16] Y.-J. Tsai, **A. Bousse**, S. Arridge, C. W. Stearns, B. F. Hutton, and K. Thielemans, “Penalized PET/CT reconstruction algorithms with automatic realignment for anatomical priors,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 5, no. 3, pp. 362–372, 2021. DOI: [10.1109/TRPMS.2020.3025540](https://doi.org/10.1109/TRPMS.2020.3025540). [Online]. Available: <https://arxiv.org/abs/1911.08012>.

- [J17] **A. Bousse**, M. Courdurier, É. C. Émond, K. Thielemans, B. F. Hutton, P. Irarrazaval, and D. Visvikis, “PET reconstruction with non-negativity constraints in projection space: Optimization through hypo-convergence,” *IEEE Transactions on Medical Imaging*, vol. 39, no. 1, pp. 75–86, 2020. DOI: [10.1109/TMI.2019.2920109](https://doi.org/10.1109/TMI.2019.2920109). [Online]. Available: <https://hal.archives-ouvertes.fr/hal-02144923>.
- [J18] L. Brusaferrri, **A. Bousse**, É. C. Émond, R. Brown, Y.-J. Tsai, D. Atkinson, S. Ourselin, C. C. Watson, B. F. Hutton, S. Arridge, *et al.*, “Joint activity, attenuation and scatter estimation from multiple energy window data in non-TOF 3D PET: A preliminary study,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 4, no. 4, pp. 410–421, 2020. DOI: [10.1109/TRPMS.2020.2978449](https://doi.org/10.1109/TRPMS.2020.2978449). [Online]. Available: <https://ieeexplore.ieee.org/document/9024002>.
- [J19] É. C. Émond, **A. Bousse**, L. Brusaferrri, B. F. Hutton, and K. Thielemans, “Improved PET/CT respiratory motion compensation by incorporating changes in lung density,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 4, no. 5, pp. 594–602, 2020. DOI: [10.1109/TRPMS.2020.3001094](https://doi.org/10.1109/TRPMS.2020.3001094). [Online]. Available: <https://ieeexplore.ieee.org/document/9112356>.
- [J20] É. C. Émond, **A. Bousse**, M. Machado, J. Porter, A. M. Groves, B. F. Hutton, and K. Thielemans, “Effect of attenuation mismatches in time of flight PET reconstruction,” *Physics in Medicine & Biology*, vol. 65, no. 8, p. 085 009, 2020. DOI: [10.1088/1361-6560/ab7a6f](https://doi.org/10.1088/1361-6560/ab7a6f). [Online]. Available: <https://iopscience.iop.org/article/10.1088/1361-6560/ab7a6f>.
- [J21] Y.-J. Tsai, G. Schramm, S. Ahn, **A. Bousse**, S. Arridge, J. Nuyts, B. F. Hutton, C. W. Stearns, and K. Thielemans, “Benefits of using a spatially-variant penalty strength with anatomical priors in PET reconstruction,” *IEEE Transactions on Medical Imaging*, vol. 39, no. 1, pp. 11–22, 2020. DOI: [10.1109/TMI.2019.2913889](https://doi.org/10.1109/TMI.2019.2913889).
- [J22] A. Iborra, A. J. González, A. Gonzalez-Montoro, **A. Bousse**, and D. Visvikis, “Ensemble of neural networks for 3D position estimation in monolithic PET detectors,” *Physics in Medicine & Biology*, vol. 64, no. 19, p. 195 010, 2019. DOI: [10.1088/1361-6560/ab3b86](https://doi.org/10.1088/1361-6560/ab3b86).
- [J23] Y.-J. Tsai, **A. Bousse**, M. J. Ehrhardt, C. W. Stearns, S. Ahn, B. F. Hutton, S. Arridge, and K. Thielemans, “Fast quasi-newton algorithms for penalized reconstruction in emission tomography and further improvements via preconditioning,” *IEEE Transactions on Medical Imaging*, vol. 37, no. 4, pp. 1000–1010, 2018. DOI: [10.1109/TMI.2017.2786865](https://doi.org/10.1109/TMI.2017.2786865). [Online]. Available: <https://doi.org/10.1109/TMI.2017.2786865>.
- [J24] **A. Bousse**, R. Manber, B. F. Holman, D. Atkinson, S. Arridge, S. Ourselin, B. F. Hutton, and K. Thielemans, “Evaluation of a direct motion estimation/correction method in respiratory-gated PET/MRI with motion-adjusted attenuation,” *Medical Physics*, vol. 44, no. 6, pp. 2379–2390, 2017. DOI: [10.1002/mp.12253](https://doi.org/10.1002/mp.12253). [Online]. Available: <https://doi.org/10.1002/mp.12253>.
- [J25] **A. Bousse**, O. Bertolli, D. Atkinson, S. Arridge, S. Ourselin, B. F. Hutton, and K. Thielemans, “Maximum-likelihood joint image reconstruction and motion estimation with misaligned attenuation in TOF-PET/CT,” *Physics in Medicine & Biology*, vol. 61, no. 3, pp. L11–19, 2016. DOI: [10.1088/0031-9155/61/3/L11](https://doi.org/10.1088/0031-9155/61/3/L11). [Online]. Available: <https://doi.org/10.1088/0031-9155/61/3/L11>.
- [J26] **A. Bousse**, O. Bertolli, D. Atkinson, S. Arridge, S. Ourselin, B. F. Hutton, and K. Thielemans, “Maximum-likelihood joint image reconstruction/motion estimation in attenuation-corrected respiratory gated PET/CT using a single attenuation map,” *IEEE Transactions on Medical Imaging*, vol. 35, no. 1, pp. 217–228, 2016. DOI: [10.1109/TMI.2015.2464156](https://doi.org/10.1109/TMI.2015.2464156). [Online]. Available: <https://doi.org/10.1109/TMI.2015.2464156>.
- [J27] J. Jiao, **A. Bousse**, K. Thielemans, N. Burgos, P. S. J. Weston, J. M. Schott, D. Atkinson, S. R. Arridge, B. F. Hutton, P. Markiewicz, and S. Ourselin, “Direct parametric reconstruction with joint motion estimation/correction for dynamic brain PET data,” *IEEE Transactions on Medical Imaging*, vol. 36, no. 1, pp. 203–213, 2016. DOI: [10.1109/TMI.2016.2594150](https://doi.org/10.1109/TMI.2016.2594150). [Online]. Available: <https://doi.org/10.1109/TMI.2016.2594150>.

- [J28] B. A. Thomas, V. Cuplov, **A. Bousse**, A. Mendes, K. Thielemans, B. F. Hutton, and K. Erlandsson, “PETPVC: a toolbox for performing partial volume correction techniques in positron emission tomography,” *Physics in Medicine & Biology*, vol. 61, no. 22, pp. 7975–7993, 2016. DOI: [10.1088/0031-9155/61/22/7975](https://doi.org/10.1088/0031-9155/61/22/7975). [Online]. Available: <https://discovery.ucl.ac.uk/id/eprint/1523346>.
- [J29] D. Salvado, K. Erlandsson, **A. Bousse**, M. Occhipinti, P. Busca, C. Fiorini, and B. F. Hutton, “Collimator design for a brain SPECT/MRI insert,” *IEEE Transactions on Nuclear Science*, vol. 62, no. 4, pp. 1716–1724, 2015. DOI: [10.1109/TNS.2015.2450017](https://doi.org/10.1109/TNS.2015.2450017). [Online]. Available: <https://doi.org/10.1109/TNS.2015.2450017>.
- [J30] B. A. Thomas, K. Erlandsson, I. Drobnjak, S. Pedemonte, K. Vunckx, **A. Bousse**, A. Reilhac-Laborde, S. Ourselin, and B. F. Hutton, “Framework for the construction of a monte carlo simulated brain PET-MR image database,” *Nuclear Instruments and Methods in Physics Research Section A*, vol. 734, pp. 162–165, 2014. DOI: [10.1016/j.nima.2013.08.063](https://doi.org/10.1016/j.nima.2013.08.063).
- [J31] B. F. Hutton, B. A. Thomas, K. Erlandsson, **A. Bousse**, A. Reilhac-Laborde, D. Kazantsev, S. Pedemonte, K. Vunckx, S. Arridge, and S. Ourselin, “What approach to brain partial volume correction is best for PET/MRI?” *Nuclear Instruments and Methods in Physics Research Section A*, vol. 702, pp. 29–33, 2013. DOI: [10.1016/j.nima.2012.07.059](https://doi.org/10.1016/j.nima.2012.07.059). [Online]. Available: <https://doi.org/10.1016/j.nima.2012.07.059>.
- [J32] **A. Bousse**, S. Pedemonte, B. A. Thomas, K. Erlandsson, S. Ourselin, S. Arridge, and B. F. Hutton, “Markov random field and gaussian mixture for segmented MRI-based partial volume correction in PET,” *Physics in Medicine & Biology*, vol. 57, no. 20, pp. 6681–6705, 2012. DOI: [10.1088/0031-9155/57/20/6681](https://doi.org/10.1088/0031-9155/57/20/6681).
- [J33] D. Kazantsev, S. Arridge, S. Pedemonte, **A. Bousse**, K. Erlandsson, B. F. Hutton, and S. Ourselin, “An anatomically driven anisotropic diffusion filtering method for 3D SPECT reconstruction,” *Physics in Medicine & Biology*, vol. 57, no. 12, p. 3793, 2012. DOI: [10.1088/0031-9155/57/12/3793](https://doi.org/10.1088/0031-9155/57/12/3793).
- [J34] **A. Bousse**, J. Zhou, G. Yang, J.-J. Bellanger, and C. Toumoulin, “Motion compensated tomography reconstruction of coronary arteries in rotational angiography,” *IEEE Transactions on Biomedical Engineering*, vol. 56, no. 4, pp. 1254–1257, 2009. DOI: [10.1109/TBME.2008.2005205](https://doi.org/10.1109/TBME.2008.2005205). [Online]. Available: <https://hal.archives-ouvertes.fr/inserm-00418315>.
- [J35] J. Zhou, J.-L. Coatrieux, **A. Bousse**, H. Shu, and L. Luo, “A bayesian MAP-EM algorithm for PET image reconstruction using wavelet transform,” *IEEE Transactions on Nuclear Science*, vol. 54, no. 5, pp. 1660–1669, 2007. DOI: [10.1109/TNS.2007.901200](https://doi.org/10.1109/TNS.2007.901200). [Online]. Available: <https://hal.archives-ouvertes.fr/inserm-00184255>.
- [J36] **A. Bousse**, C. Boldak, C. Toumoulin, G. Yang, S. Laguitton, and D. Boulmier, “Coronary extraction and characterization in multi-detector computed tomography,” *ITBM-RBM*, vol. 27, no. 5, pp. 217–226, 2006. DOI: [10.1016/j.rbmret.2007.01.001](https://doi.org/10.1016/j.rbmret.2007.01.001). [Online]. Available: <https://doi.org/10.1016/j.rbmret.2007.01.001>.

Review Articles

- [RV1] **A. Bousse**, V. S. S. Kandarpa, S. Rit, A. Perelli, M. Li, G. Wang, J. Zhou, and G. Wang, “Systematic review on learning-based spectral CT,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 8, no. 2, pp. 113–137, 2024. DOI: [10.1109/TRPMS.2023.3314131](https://doi.org/10.1109/TRPMS.2023.3314131). [Online]. Available: <https://arxiv.org/abs/2304.07588>.
- [RV2] **A. Bousse**, V. S. S. Kandarpa, K. Shi, K. Gong, J. S. Lee, C. Liu, and D. Visvikis, “A review on low-dose emission tomography post-reconstruction denoising with neural network approaches,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 8, no. 4, pp. 333–347, 2024. DOI: [10.1109/TRPMS.2023.3349194](https://doi.org/10.1109/TRPMS.2023.3349194). [Online]. Available: <https://arxiv.org/abs/2401.00232>.

- [RV3] F. Lamare, **A. Bousse**, K. Thielemans, C. Liu, T. Merlin, H. Fayad, and D. Visvikis, “PET respiratory motion correction: Quo vadis?” *Physics in Medicine & Biology*, vol. 67, no. 3, 03TR02, 2022. DOI: [10.1088/1361-6560/ac43fc](https://doi.org/10.1088/1361-6560/ac43fc). [Online]. Available: <https://hal.science/hal-04090024>.

Preprints

- [PP1] C. Phung-Ngoc, **A. Bousse**, A. De Paepe, H.-P. Dang, O. Saut, and D. Visvikis, “Joint reconstruction of activity and attenuation in PET by diffusion posterior sampling in wavelet coefficient space,” *arXiv preprint arXiv:2505.18782*, 2025. [Online]. Available: <https://arxiv.org/abs/2505.18782>.
- [PP2] H. Xu, **A. Bousse**, and A. Perelli, “Direct dual-energy CT material decomposition using model-based denoising diffusion model,” *arXiv preprint arXiv:2507.18012*, 2025. [Online]. Available: <https://arxiv.org/abs/2507.18012>.
- [PP3] J. Molina, **A. Bousse**, T. Catalán, Z. Wang, M. Petrache, F. Sahli, C. Prieto, and M. Courdurier, “CConnect: Synergistic convolutional regularization for cartesian T2* mapping,” *arXiv preprint arXiv:2404.18182*, 2024. [Online]. Available: <https://arxiv.org/abs/2404.18182>.

Conference Proceedings and Presentations

- [CP1] T. Dassow, C. Vazia, A. Perelli, D. Visvikis, and **A. Bousse**, “Out-of-database diffusion posterior sampling for spectral CT material decomposition,” in *International Conference on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine*, 2025.
- [CP2] A. De Paepe, **A. Bousse**, C. Phung-Ngoc, and D. Visvikis, “Solving blind inverse problems: Adaptive diffusion models for motion-corrected sparse-view 4DCT,” in *International Conference on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine*, 2025. [Online]. Available: <https://arxiv.org/abs/2501.12249>.
- [CP3] C. Phung-Ngoc, **A. Bousse**, A. De Paepe, H.-P. Dang, O. Saut, and D. Visvikis, “Joint reconstruction of the activity and the attenuation in PET by diffusion posterior sampling: A feasibility study,” in *International Conference on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine*, 2025. [Online]. Available: <https://arxiv.org/abs/2412.11776>.
- [CP4] Z. Wang, **A. Bousse**, F. Vermet, J. Froment, B. Vedel, A. Perelli, and D. Visvikis, “MHUconnect: Multi-head U-Net connecting all energy bins for synergistic spectral CT reconstruction,” in *International Conference on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine*, 2025.
- [CP5] V. Gautier, C. Comtat, F. Sureau, **A. Bousse**, V. Maxim, and B. Sixou, “Synergistic PET/MR reconstruction with VAE constraint,” in *EUSIPCO 2024-32nd European Signal Processing Conference*, 2024. DOI: [10.23919/EUSIPCO63174.2024.10715319](https://doi.org/10.23919/EUSIPCO63174.2024.10715319). [Online]. Available: <https://hal.science/hal-04754695/document>.
- [CP6] Y. Mellak, **A. Bousse**, T. Merlin, É. C. Émond, and D. Visvikis, “One linear layer is all you need for positron range estimation and correction,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Detector Conference*, IEEE, 2024, pp. 1–1.
- [CP7] Y. Mellak, **A. Bousse**, T. Merlin, B. Laurent, and D. Visvikis, “Detector-pixel interactions via transformer encoders: A direct approach for PET image reconstruction,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Detector Conference*, IEEE, 2024, pp. 1–1.

- [CP8] Y. Mellak, K. Chatzipapas, **A. Bousse**, C. Chez-Le Rest, D. Visvikis, and J. Bert, “Fast-track of F-18 positron paths simulations using GANs,” in *2024 IEEE 21st international symposium on biomedical imaging (ISBI 2024)*, IEEE, 2024. [Online]. Available: <https://arxiv.org/abs/2403.06307>.
- [CP9] C. Vazia, **A. Bousse**, J. Froment, B. Vedel, F. Vermet, Z. Wang, T. Dassow, J.-P. Tasu, and D. Visvikis, “Spectral CT two-step and one-step material decomposition using diffusion posterior sampling,” in *European Signal Processing Conference (EUSIPCO)*, 2024. doi: [10.23919/EUSIPCO63174.2024.10715152](https://arxiv.org/abs/2403.10183). [Online]. Available: <https://arxiv.org/abs/2403.10183>.
- [CP10] C. Vazia, **A. Bousse**, B. Vedel, F. Vermet, Z. Wang, T. Dassow, J.-P. Tasu, D. Visvikis, and J. Froment, “Diffusion posterior sampling for synergistic reconstruction in spectral computed tomography,” in *2024 IEEE 21st international symposium on biomedical imaging (ISBI 2024)*. IEEE, 2024. [Online]. Available: <https://arxiv.org/abs/2403.06308>.
- [CP11] J. Zhang, **A. Bousse**, Y. Li, S. Xue, K. Shi, D. Visvikis, and J. Bert, “Pre-therapy dose prediction in targeted radionuclide therapy using semi-supervised learning: An in-silico preliminary study,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Detector Conference*, IEEE, 2024, pp. 1–1.
- [CP12] V. Gautier, C. Comtat, F. Sureau, **A. Bousse**, L. Friot-Giroux, V. Maxim, and B. Sixou, “VAE constrained MR guided PET reconstruction,” in *International Conference on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine*, 2023.
- [CP13] N. J. Pinton, **A. Bousse**, C. Cheze-Le Rest, and D. Visvikis, “Joint PET/CT reconstruction using a double variational autoencoder,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2023.
- [CP14] N. J. Pinton, **A. Bousse**, Z. Wang, C. Cheze-Le Rest, V. Maxim, C. Comtat, F. Sureau, and D. Visvikis, “Synergistic PET/CT reconstruction using a joint generative model,” in *International Conference on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine*, 2023.
- [CP15] V. S. S. Kandarpa, **A. Bousse**, T. Merlin, A. Perelli, S. Xue, R. Ma, Y. Li, H. Sari, A. Rominger, K. Shi, and D. Visvikis, “C-DenseNet: A direct neural network-based approach for total body PET image reconstruction,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2022.
- [CP16] Y. Mellak, D. Giovagnoli, **A. Bousse**, and D. Visvikis, “Three-gamma PET image reconstruction using graph neural networks,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2022.
- [CP17] Z. Wang, **A. Bousse**, F. Vermet, N. J. Pinton, J. Froment, B. Vedel, J.-P. Tasu, and D. Visvikis, “Synergistic multi-energy reconstruction with a deep penalty “connecting the energies”,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2022.
- [CP18] S. L. Alfonso Garcia, A. Perelli, **A. Bousse**, and D. Visvikis, “Dual-energy CT reconstruction with convolutional analysis operator learning,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2021.
- [CP19] S. L. Alfonso Garcia, A. Perelli, **A. Bousse**, and D. Visvikis, “Sparse-view joint reconstruction and material decomposition for dual-energy cone-beam computed tomography,” in *International Conference on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine*, 2021, pp. 389–394. [Online]. Available: <https://arxiv.org/abs/2110.04143>.
- [CP20] V. S. S. Kandarpa, A. Perelli, **A. Bousse**, and D. Visvikis, “Deep learning based direct sparse-view CT image reconstruction with concatenated U-Net,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2021.

- [CP21] S. L. Alfonso Garcia, **A. Bousse**, and D. Visvikis, “A coupled image-motion dictionary learning algorithm for motion estimation-compensation in cone-beam computed tomography,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2020.
- [CP22] L. Brusafferri, É. C. Émond, **A. Bousse**, R. Twyman, D. Atkinson, B. F. Hutton, S. Arridge, and K. Thielemans, “Normalisation factor estimation in non-TOF 3D PET from multiple-energy window data,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2020.
- [CP23] B. Laurent, T. Merlin, **A. Bousse**, and D. Visvikis, “Deep learning based scatter correction for PET imaging,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2020.
- [CP24] B. Le Crom, **A. Bousse**, M. Chérel, N. Costes, S. Gouard, S. Marionneau-Lambot, T. Merlin, D. Visvikis, S. Stute, and T. Carlier, “A single dual-tracer PET imaging acquisition to provide information on tumor heterogeneities,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2020.
- [CP25] É. C. Emond, **A. Bousse**, M. Machado, J. Porter, K. Erlandsson, A. M. Groves, B. F. Hutton, and K. Thielemans, “Respiratory motion correction in dynamic PET with a single attenuation map,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, IEEE, 2019, pp. 1–3.
- [CP26] É. C. Émond, **A. Bousse**, L. Brusafferri, A. M. Groves, B. F. Hutton, and K. Thielemans, “Mass preservation for respiratory motion registration in both PET and CT,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2019.
- [CP27] É. C. Émond, **A. Bousse**, A. M. Groves, B. F. Hutton, and K. Thielemans, “Dependence of error propagation due to an incorrect attenuation map on PET time-of-flight resolution,” in *Eur. Assoc. Nucl. Med.*, 2019.
- [CP28] A. Iborra, J. González, A. González-Montoro, **A. Bousse**, and D. Visvikis, “Ensemble of neural networks for 3D position estimation in monolithic PET detectors,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2019.
- [CP29] V. S. S. Kandarpa, D. Benoit, **A. Bousse**, and D. Visvikis, “Direct image reconstruction using generative deep learning networks,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2019.
- [CP30] **A. Bousse**, M. Courdurier, É. C. Émond, K. Thielemans, B. F. Hutton, P. Irarrazaval, and D. Visvikis, “PET reconstruction with non-negativity constraints in projection space: Optimization through hypo-convergence,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2018.
- [CP31] **A. Bousse**, S. Pedemonte, B. A. Thomas, K. Erlandsson, S. Ourselin, S. Arridge, and B. F. Hutton, “Markov random field and gaussian mixture for segmented MRI-based partial volume correction in PET,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2018.
- [CP32] L. Brusafferri, **A. Bousse**, Y.-J. Tsai, D. Atkinson, S. Ourselin, B. F. Hutton, S. Arridge, and K. Thielemans, “Maximum-likelihood estimation of emission and attenuation images in 3D PET from multiple energy window measurements,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2018. DOI: [10.1109/NSSMIC.2018.8824557](https://doi.org/10.1109/NSSMIC.2018.8824557).

- [CP33] Y.-J. Tsai, **A. Bousse**, S. Ahn, C. W. Stearns, S. Arridge, B. F. Hutton, and K. Thielemans, "Algorithms for solving misalignment issues in penalized PET/CT reconstruction using anatomical priors," in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2018. DOI: [10.1109/NSSMIC.2018.8824558](https://doi.org/10.1109/NSSMIC.2018.8824558).
- [CP34] **A. Bousse**, B. F. Hutton, and K. Thielemans, "Fast gated PET direct motion estimation using ordered subsets," in *International Conference on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine*, 2017. DOI: [10.12059/Fully3D.2017-11-3202020](https://doi.org/10.12059/Fully3D.2017-11-3202020). [Online]. Available: <http://onlinelibrary.fully3d.org/papers/2017/Fully3D.2017-11-3202020.pdf>.
- [CP35] L. Brusaferri, **A. Bousse**, N. Efthimiou, É. C. Émond, D. Atkinson, S. Ourselin, B. F. Hutton, S. Arridge, and K. Thielemans, "Potential benefits of incorporating energy information when estimating attenuation from PET data," in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2017. DOI: [10.1109/NSSMIC.2017.8532765](https://doi.org/10.1109/NSSMIC.2017.8532765).
- [CP36] Y.-J. Tsai, G. Schramm, J. Nuyts, S. Ahn, C. W. Stearns, **A. Bousse**, S. Arridge, and K. Thielemans, "Spatially-variant strength for anatomical priors in PET reconstruction," in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2017. DOI: [10.1109/NSSMIC.2017.8532925](https://doi.org/10.1109/NSSMIC.2017.8532925).
- [CP37] **A. Bousse**, A. Sidlesky, N. Roth, A. Rashidnasab, K. Thielemans, and B. F. Hutton, "Joint activity/attenuation reconstruction in SPECT using photopeak and scatter sinograms," in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2016. DOI: [10.1109/NSSMIC.2016.8069448](https://doi.org/10.1109/NSSMIC.2016.8069448).
- [CP38] A. Rashidnasab, **A. Bousse**, B. F. Holman, B. F. Hutton, and K. Thielemans, "Joint reconstruction of activity and attenuation in dynamic PET," in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2016. DOI: [10.1109/NSSMIC.2016.8069456](https://doi.org/10.1109/NSSMIC.2016.8069456).
- [CP39] Y.-J. Tsai, **A. Bousse**, C. W. Stearns, S. Ahn, B. F. Hutton, S. Arridge, and K. Thielemans, "Performance improvement and validation of a new MAP reconstruction algorithm," in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2016. DOI: [10.1109/NSSMIC.2016.8069458](https://doi.org/10.1109/NSSMIC.2016.8069458).
- [CP40] **A. Bousse**, O. Bertolli, D. Atkinson, S. Arridge, S. Ourselin, B. F. Hutton, and K. Thielemans, "Direct joint motion estimation/image reconstruction in attenuation-corrected gated PET/CT without gated CT," in *International Conference on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine*, 2015.
- [CP41] K. Erlandsson, D. Salvado, **A. Bousse**, and B. F. Hutton, "Evaluation of a partial ring design for the INSERT SPECT/MRI system," in *EJNMMI physics*, vol. 2, Springer, 2015, A47. DOI: [10.1186/2197-7364-2-S1-A47](https://doi.org/10.1186/2197-7364-2-S1-A47).
- [CP42] Y.-J. Tsai, **A. Bousse**, M. J. Ehrhardt, B. F. Hutton, S. Arridge, and K. Thielemans, "Performance evaluation of MAP algorithms with different penalties, object geometries and noise levels," in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2015. DOI: [10.1109/NSSMIC.2015.7582101](https://doi.org/10.1109/NSSMIC.2015.7582101).
- [CP43] **A. Bousse**, J. Jiao, K. Erlandsson, L. Pizarro, K. Thielemans, D. Atkinson, S. Ourselin, S. Arridge, and B. F. Hutton, "4-D PET joint image reconstruction/non-rigid motion estimation with limited MRI prior information," in *EJNMMI physics*, vol. 1, Springer, 2014, A27. DOI: [10.1186/2197-7364-1-S1-A27](https://doi.org/10.1186/2197-7364-1-S1-A27).
- [CP44] **A. Bousse**, J. Jiao, L. Pizarro, K. Thielemans, D. Atkinson, S. Ourselin, S. Arridge, and B. F. Hutton, "An algorithm for direct 4-D PET image reconstruction/non-rigid motion estimation with limited MRI prior information," in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2014, pp. 1–3. DOI: [10.1109/NSSMIC.2014.7430828](https://doi.org/10.1109/NSSMIC.2014.7430828).

- [CP45] J. Jiao, **A. Bousse**, K. Thielemans, P. Markiewicz, N. Burgos, D. Atkinson, S. Arridge, B. F. Hutton, and S. Ourselin, “Joint parametric reconstruction and motion correction framework for dynamic PET data,” in *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2014: 17th International Conference, Boston, MA, USA, September 14–18, 2014, Proceedings, Part I 17*, Springer, 2014, pp. 114–121. DOI: [10.1007/978-3-319-10404-1_15](https://doi.org/10.1007/978-3-319-10404-1_15).
- [CP46] D. Salvado, K. Erlandsson, **A. Bousse**, M. Occhipinti, P. Busca, C. Fiorini, B. F. Hutton, *et al.*, “Collimator design for a clinical brain SPECT/MRI insert,” in *EJNMMI physics*, vol. 1, Springer, 2014, A21. DOI: [10.1186/2197-7364-1-S1-A21](https://doi.org/10.1186/2197-7364-1-S1-A21). [Online]. Available: <https://doi.org/10.1186/2197-7364-1-S1-A21>.
- [CP47] **A. Bousse**, K. Erlandsson, N. Fuin, D. Salvado, and B. F. Hutton, “Variance prediction in SPECT reconstruction based on the fisher information using a novel angular blurring algorithm for computation of the system matrix,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2013, pp. 1–6. DOI: [10.1109/NSSMIC.2013.6829221](https://doi.org/10.1109/NSSMIC.2013.6829221).
- [CP48] **A. Bousse**, K. Erlandsson, S. Pedemonte, S. Ourselin, S. Arridge, and B. F. Hutton, “Angular rebinning for geometry independent SPECT reconstruction,” in *International Conference on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine*, 2013.
- [CP49] K. Erlandsson, D. Salvado, **A. Bousse**, and B. F. Hutton, “Design optimization and evaluation of a human brain SPECT-MRI insert based on high-resolution detectors and slit-slat collimators,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2013, pp. 1–4. DOI: [10.1109/NSSMIC.2013.6829144](https://doi.org/10.1109/NSSMIC.2013.6829144).
- [CP50] **A. Bousse**, C. Panagiotou, K. Erlandsson, S. Ourselin, S. Arridge, and B. F. Hutton, “Monotonic algorithm for joint entropy-based anatomical priors in parametric PET image reconstruction,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2012, pp. 3918–3924. DOI: [10.1109/NSSMIC.2012.6551899](https://doi.org/10.1109/NSSMIC.2012.6551899).
- [CP51] B. A. Thomas, K. Erlandsson, A. Reilhac, **A. Bousse**, D. Kazantsev, S. Pedemonte, K. Vunckx, S. Arridge, S. Ourselin, and B. F. Hutton, “A comparison of the options for brain partial volume correction using PET/MRI,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2012, pp. 2902–2906. DOI: [10.1109/NSSMIC.2012.6551662](https://doi.org/10.1109/NSSMIC.2012.6551662).
- [CP52] K. Vunckx, S. Arridge, **A. Bousse**, D. Kazantsev, S. Pedemonte, S. Ourselin, and B. F. Hutton, “Unifying global and local statistical measures for anatomy-guided emission tomography reconstruction,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2012, pp. 2161–2165. DOI: [10.1109/NSSMIC.2012.6551494](https://doi.org/10.1109/NSSMIC.2012.6551494).
- [CP53] D. Kazantsev, S. Arridge, S. Pedemonte, S. Ourselin, **A. Bousse**, and B. F. Hutton, “Robust anisotropic diffusion prior with anatomical regularization for 3D SPECT reconstruction,” in *International Conference on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine*, 2011.
- [CP54] D. Kazantsev, **A. Bousse**, S. Pedemonte, S. Arridge, B. F. Hutton, and S. Ourselin, “Edge preserving Bowsher prior with nonlocal weighting for 3D SPECT reconstruction,” in *IEEE Int. Symp. on Bio-med. Imag.: From Nano to Macro*, 2011, pp. 1158–1161. DOI: [10.1109/ISBI.2011.5872607](https://doi.org/10.1109/ISBI.2011.5872607).
- [CP55] S. Pedemonte, **A. Bousse**, B. F. Hutton, S. Arridge, and S. Ourselin, “4-D generative model for PET/MRI reconstruction,” in *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2011: 14th International Conference, Toronto, Canada, September 18–22, 2011, Proceedings, Part I 14*, Springer, 2011, pp. 581–588. DOI: [10.1007/978-3-642-23623-5_73](https://doi.org/10.1007/978-3-642-23623-5_73).

- [CP56] S. Pedemonte, **A. Bousse**, B. F. Hutton, S. Arridge, and S. Ourselin, “Probabilistic graphical model of SPECT/MRI,” in *Machine Learning in Med. Imag.*, 2011, pp. 167–174. DOI: [10.1007/978-3-642-24319-6_21](https://doi.org/10.1007/978-3-642-24319-6_21).
- [CP57] **A. Bousse**, S. Pedemonte, D. Kazantsev, S. Ourselin, S. Arridge, and B. F. Hutton, “Weighted MRI-based Bowsher priors for SPECT brain image reconstruction,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2010, pp. 3519–3522. DOI: [10.1109/NSSMIC.2010.5874462](https://doi.org/10.1109/NSSMIC.2010.5874462).
- [CP58] S. Cade, **A. Bousse**, S. Arridge, M. Evans, and B. F. Hutton, “Estimating an attenuation map from measured scatter for 180o cardiac SPECT,” in *Soc. of Nuclear. Med. Abstracts*, vol. 51, 2010, p. 1357.
- [CP59] D. Kazantsev, S. Pedemonte, **A. Bousse**, C. Panagiotou, S. Arridge, B. F. Hutton, and S. Ourselin, “ET bayesian reconstruction using automatic bandwidth selection for joint entropy optimization,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2010, pp. 3301–3307. DOI: [10.1109/NSSMIC.2010.5874415](https://doi.org/10.1109/NSSMIC.2010.5874415).
- [CP60] S. Pedemonte, **A. Bousse**, K. Erlandsson, M. Modat, S. Arridge, B. F. Hutton, and S. Ourselin, “GPU accelerated rotation-based emission tomography reconstruction,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2010, pp. 2657–2661. DOI: [10.1109/NSSMIC.2010.5874272](https://doi.org/10.1109/NSSMIC.2010.5874272).
- [CP61] S. Pedemonte, M. J. Cardoso, **A. Bousse**, C. Panagiotou, D. Kazantsev, S. Arridge, B. F. Hutton, and S. Ourselin, “Class conditional entropic prior for MRI enhanced SPECT reconstruction,” in *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detectors Conference*, 2010, pp. 3292–3300. DOI: [10.1109/NSSMIC.2010.5874414](https://doi.org/10.1109/NSSMIC.2010.5874414).
- [CP62] **A. Bousse**, J. Zhou, G. Yang, J.-J. Bellanger, and C. Toumoulin, “Motion estimation in x-ray rotational angiography using a 3-D deformable coronary tree model,” in *Comp. in Cardio.*, 2008, pp. 529–532. DOI: [10.1109/CIC.2008.4749095](https://doi.org/10.1109/CIC.2008.4749095).
- [CP63] J. Zhou, **A. Bousse**, G. Yang, J.-J. Bellanger, L. Luo, C. Toumoulin, and J.-L. Coatrieux, “A blob-based tomographic reconstruction of 3D coronary trees from rotational X-ray angiography,” in *Medical Imaging 2008: Physics of Medical Imaging*, International Society for Optics and Photonics, vol. 6913, 2008, pp. 886–897. DOI: [10.1117/12.769478](https://doi.org/10.1117/12.769478). [Online]. Available: <https://hal.archives-ouvertes.fr/inserm-00335244>.
- [CP64] G. Yang, **A. Bousse**, C. Toumoulin, and H. Shu, “Simulation environment for the evaluation of 3D coronary tree reconstruction algorithms in rotational angiography,” in *International Conference of the IEEE Engineering in Medicine and Biology Society*, 2007, pp. 4484–4487. DOI: [10.1109/IEMBS.2007.4353335](https://doi.org/10.1109/IEMBS.2007.4353335).
- [CP65] S. Laguitton, C. Boldak, **A. Bousse**, G. Yang, and C. Toumoulin, “Temporal tracking of coronaries in MSCTA by means of 3D geometrical moments,” in *International Conference of the IEEE Engineering in Medicine and Biology Society*, 2006, pp. 924–927. DOI: [10.1109/IEMBS.2006.260670](https://doi.org/10.1109/IEMBS.2006.260670).
- [CP66] G. Yang, **A. Bousse**, C. Toumoulin, and H. Shu, “A multiscale tracking algorithm for the coronary extraction in MSCT angiography,” in *International Conference of the IEEE Engineering in Medicine and Biology Society*, 2006, pp. 3066–3069. DOI: [10.1109/IEMBS.2006.260712](https://doi.org/10.1109/IEMBS.2006.260712).
- [CP67] J. Brieva, E. M. Á. González, F. González, **A. Bousse**, and J.-J. Bellanger, “A level set method for vessel segmentation in coronary angiography,” in *International Conference of the IEEE Engineering in Medicine and Biology Society*, vol. 6, 2005, pp. 6348–6351. DOI: [10.1109/IEMBS.2005.1615949](https://doi.org/10.1109/IEMBS.2005.1615949).